UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

DECEMBER EXAMINATION

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES

WEDNESDAY, 21 DECEMBER 2022 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instructions to Candidates :

This question paper consists of FIVE (5) questions .

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

Candidates can find formulae in the appendix.

Section A (Answer All Questions)

Q1. Given student B attendance in a class is 80%, while student T attendance is 60%.

(a) What is the probability for both students to be absent from a class? (4 marks)

(b) What is the probability for at least one of the students to attend the class on any given day? (5 marks)

(c) If there is student S who always attends the class if and only if student B or student T or both show up.

(i) Construct the Bayes Network topology to show the relationship between the three students: B, T, S. Include the conditional probability tables in the topology. (4 marks)

(ii) Based on the constructed Bayes Network topology in Q1(c)(i), analyze whether student B is independent of T given S. (3 marks)

(iii) Deriving from the Bayes rule, if both students S and T have come to a class, what is the probability of student B coming to the class too? (9 marks)

[Total : 25 marks]

Q2. Given the following Table 2.1 dataset:

Table 2.1

Dataset	Actual	Predict
1	😷	🙂
2	🙂	🙂
3	😷	😷
4	🙂	😷
5	😷	🙂
6	🙂	🙂
7	😷	😷
8	🙂	🙂
9	😷	😷
10	😷	😷

Q2. (Continued)

(a) Draw the confusion matrix to represent Table 2.1 data. Consider emoji with mask is positive class, and without mask is negative class. (3 marks)

(b) Based on the confusion matrix you have drawn in Q2(a), calculate:

(i) Accuracy (3 marks)

(ii) Precision (3 marks)

(iii) Recall (3 marks)

(iv) F1-score (3 marks)

(c) A data scientist is working on optimizing a model during the training process by varying multiple parameters. The data scientist observes that, during multiple runs with identical parameters, the loss function converges to different, yet stable, values. What should the data scientist do to improve the training process? Justify your answer. (4 marks)

(d) How does splitting a dataset into training, testing, and validation sets help identifying overfitting? (6 marks)
[Total : 25 marks]

Q3. (a) Given following graph state:

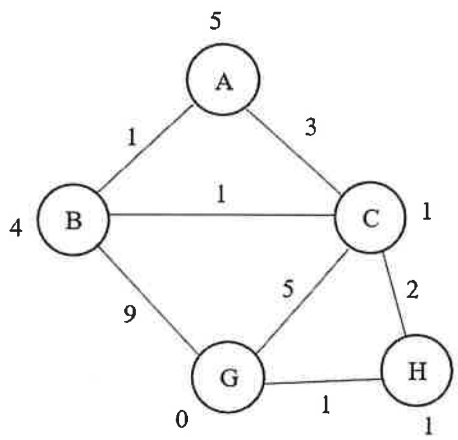

Fig. Q3

Q3.(a)(Continued)

If the goal is to reach from state A to state G, answer the following questions:

(i) Draw the A* search tree (10 marks)

(ii) What is the path and cost found by the A* search tree from A to G? (2 marks)

(iii) Is the solution optimal? Justify your answer. (2 marks)

(iv) How many nodes have been explored? (2 marks)

(b) What is the potential issue with the search tree you have drawn in Q3(a)? Justify your answer. (3 marks)

(c) What is the problem raised if admissible heuristics were used to address the potential issue faced with the A* search tree? Justify your answer. (6 marks)
[Total : 25 marks]

Section B (Choose Any ONE (1) Question)

Q4. Table 4.1 shows samples extracted from a dataset. Your task is to predict the y based on x.

Table: 4.1

Samples	x	y
1	81	609.9
2	80	536.7
3	75	323.1
4	71	237.4
5	69	266.3

(a) Assume a linear model is used to model the samples. Answer the following questions:

(i) Given two linear models A($\theta_0 = -1700, \theta_1 = 27$), and B($\theta_0 = 200, \theta_1 = 10$), determine which model is a better fit for the samples based on MSE. Show your calculation steps. (9 marks)

(ii) Gradient descent is used to find the best parameter to model the samples. Given the parameter values are initliazed to ($\theta_0 = 200, \theta_1 = 10$), compute the updated parameter values after the first iteration. Show your calculation steps. (7 marks)

Q4. (a)(Continued)

(iii) You are doing full batch gradient descent using the entire training set. Is it necessary to shuffle the training data? Explain with a short answer. (3 marks)

(b) What will go wrong if the learning rate of batch gradient descent setting is too high? (2 marks)

(c) What is the limitation of batch gradient descent? (4 marks)
[Total : 25 marks]

Q5. Table 5.1 shows sample extracted from a dataset. Your task is to classify y based on x.

Table 5.1

Samples	x	y
1	3	No
2	5	No
3	7	Yes
4	9	Yes

(a) Assume a logistic regression is used to model the samples. Answer the following questions:

(i) Given model A($\theta_0 = -3, \theta_1 = 0.5$), and B($\theta_0 = -1, \theta_1 = 0.3$), determine which model is a better fit for the samples based on accuracy scores. Show your calculation steps. (9 marks)

(ii) Gradient descent is used to find the best parameter to model the samples. Given the parameters are initliazed to ($\theta_0 = -1, \theta_1 = 0.3$), and learning rate $\eta = 0.01$ compute the updated parameter values after the first iteration. Show your calculation steps. (8 marks)

(iii) You are doing full batch gradient using the 60% of cross validation training set. Is it necessary to shuffle the training data? Explain with a short answer. (3 marks)

(b) What will go wrong if the learning rate of batch gradient descent setting is too small? (2 marks)

(c) What are the advantages of batch gradient descent? (3 marks)
[Total : 25 marks]

Appendix

$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP}$$

$$Precision = \frac{TP}{FP + TP}$$

$$F_1 = \frac{TP}{TP + \frac{FN + FP}{2}}$$

$$Recall = \frac{TP}{FN + TP}$$

$$h(\mathbf{x}) = \theta^T \mathbf{x}$$

$$\mathbf{h} = \mathbf{X} \cdot \theta$$

$$MSE(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$$RMSE(\theta) = \sqrt{MSE(\theta)}$$

$$\nabla_\theta MSE(\theta) = \frac{2}{m}\mathbf{X}^T \cdot (\mathbf{X} \cdot \theta - \mathbf{y})$$

$$P(A, B) = P(A|B)P(B)$$
$$P(A, B) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A) + P(\neg A) = 1$$